

Block 4 – Cloud Computing

S2: Cloud Computing & Cloud Architecture

Arquitectura de Xarxes

Universitat Pompeu Fabra



- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

If We Can Virtualize Everything... Why Not Sell It?

*Virtualization is the technology.
Cloud computing is the business model.*

Learning objectives:

- 1 Understand cloud computing's definition and essential characteristics

If We Can Virtualize Everything... Why Not Sell It?

*Virtualization is the technology.
Cloud computing is the business model.*

Learning objectives:

- 1 Understand cloud computing's definition and essential characteristics
- 2 Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) and identify key cloud platforms

If We Can Virtualize Everything... Why Not Sell It?

*Virtualization is the technology.
Cloud computing is the business model.*

Learning objectives:

- 1 Understand cloud computing's definition and essential characteristics
- 2 Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) and identify key cloud platforms
- 3 Design a cloud virtual network with subnets, route tables, and security controls

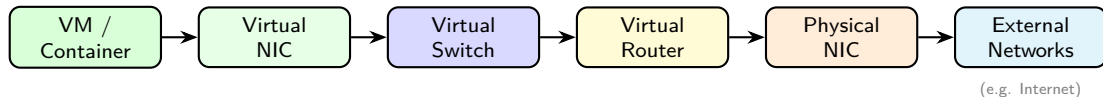
If We Can Virtualize Everything... Why Not Sell It?

*Virtualization is the technology.
Cloud computing is the business model.*

Learning objectives:

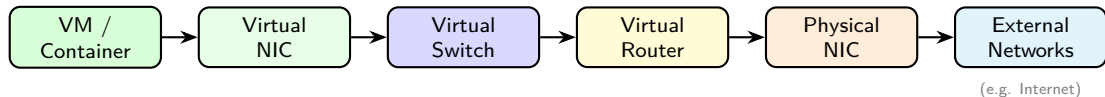
- 1 Understand cloud computing's definition and essential characteristics
- 2 Distinguish IaaS (Infrastructure as a Service), PaaS (Platform as a Service), and SaaS (Software as a Service) and identify key cloud platforms
- 3 Design a cloud virtual network with subnets, route tables, and security controls
- 4 Identify essential cloud network services: load balancers, proxies, and VPNs

Recap: What We Virtualized (Session 1)



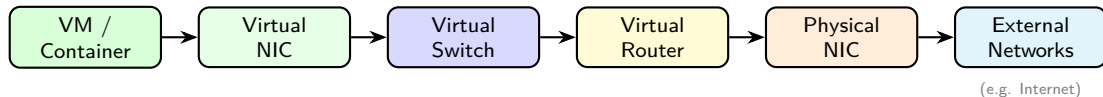
- **Compute:** VMs share physical hardware via hypervisors; containers share the host kernel directly

Recap: What We Virtualized (Session 1)



- **Compute:** VMs share physical hardware via hypervisors; containers share the host kernel directly
- **Networking:** Virtual NICs, Switches, and Routers replicate the physical stack in software

Recap: What We Virtualized (Session 1)



- **Compute:** VMs share physical hardware via hypervisors; containers share the host kernel directly
- **Networking:** Virtual NICs, Switches, and Routers replicate the physical stack in software
- **Isolation:** VLANs and overlay networks separate tenants on shared infrastructure

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	Cloud Computing
Run multiple VMs/containers on one host	Offer compute as a service

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	Cloud Computing
Run multiple VMs/containers on one host	Offer compute as a service

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	Cloud Computing
Run multiple VMs/containers on one host	Offer compute as a service
Manual provisioning possible	Automated, self-service

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

The Leap to Cloud

- Virtualization = technology (**how** you run multiple workloads)
- Cloud = business model (**what** you sell to customers)

Virtualization	Cloud Computing
Run multiple VMs/containers on one host	Offer compute as a service
Manual provisioning possible	Automated, self-service
You own the hardware	Provider owns the hardware

Key difference: **automation and self-service** (no phone calls, no tickets, no waiting).

Meet the Scenario: A Startup from Zero

Running example for this session

To understand concepts in the cloud, we will use "**CloudBite**". This is a 3-person startup that we just created and received seed funding. We need to deploy a food-delivery web app for thousands of users **by next month**.

We have laptops, code, and zero infrastructure.

Discussion: From Virtualization to Cloud

Cloudbite is running in some VMs in some spare servers. Why would you move to the cloud instead of keeping everything in-house?

Think about:

- Capital expenditure vs operational expenditure

Discussion: From Virtualization to Cloud

Cloudbite is running in some VMs in some spare servers. Why would you move to the cloud instead of keeping everything in-house?

Think about:

- Capital expenditure vs operational expenditure
- How fast can you scale today?

Discussion: From Virtualization to Cloud

Cloudbite is running in some VMs in some spare servers. Why would you move to the cloud instead of keeping everything in-house?

Think about:

- Capital expenditure vs operational expenditure
- How fast can you scale today?
- Who patches the hardware at 3 am?

Discussion: From Virtualization to Cloud

Cloudbite is running in some VMs in some spare servers. Why would you move to the cloud instead of keeping everything in-house?

Think about:

- Capital expenditure vs operational expenditure
- How fast can you scale today?
- Who patches the hardware at 3 am?
- What happens when a disk fails?

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing**
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

Cloud Computing: Definition (NIST)

NIST SP 800-145 (2011)

*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

¹NIST SP 800-145, Mell & Grance, 2011. Still the standard definition used industry-wide.

Cloud Computing: Definition (NIST)

NIST SP 800-145 (2011)

*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

Three phrases to remember:

- **On-demand:** you get resources immediately, no human approval needed

1

¹NIST SP 800-145, Mell & Grance, 2011. Still the standard definition used industry-wide.

Cloud Computing: Definition (NIST)

NIST SP 800-145 (2011)

*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

Three phrases to remember:

- **On-demand:** you get resources immediately, no human approval needed
- **Shared pool:** underlying hardware is shared across many customers (multi-tenancy)

1

¹NIST SP 800-145, Mell & Grance, 2011. Still the standard definition used industry-wide.

Cloud Computing: Definition (NIST)

NIST SP 800-145 (2011)

*"A model for enabling ubiquitous, convenient, **on-demand network access** to a **shared pool** of configurable computing resources that can be rapidly provisioned and released with **minimal management effort**."*

Three phrases to remember:

- **On-demand:** you get resources immediately, no human approval needed
- **Shared pool:** underlying hardware is shared across many customers (multi-tenancy)
- **Minimal effort:** you click a button (or call an API); the provider handles the rest

1

¹NIST SP 800-145, Mell & Grance, 2011. Still the standard definition used industry-wide.

Five Essential Characteristics

Characteristic	What it means	Real-world example
On-demand self-service	Provision via Application Programming Interface (API), no human interaction	AWS console: launch VM in 30 s

Five Essential Characteristics

Characteristic	What it means	Real-world example
On-demand self-service	Provision via Application Programming Interface (API), no human interaction	AWS console: launch VM in 30 s
Broad network access	Available from any device over the network	Access S3 from laptop, phone, VM

Five Essential Characteristics

Characteristic	What it means	Real-world example
On-demand self-service	Provision via Application Programming Interface (API), no human interaction	AWS console: launch VM in 30 s
Broad network access	Available from any device over the network	Access S3 from laptop, phone, VM
Resource pooling	Shared infrastructure, multi-tenant	Same physical server, different tenants

Five Essential Characteristics

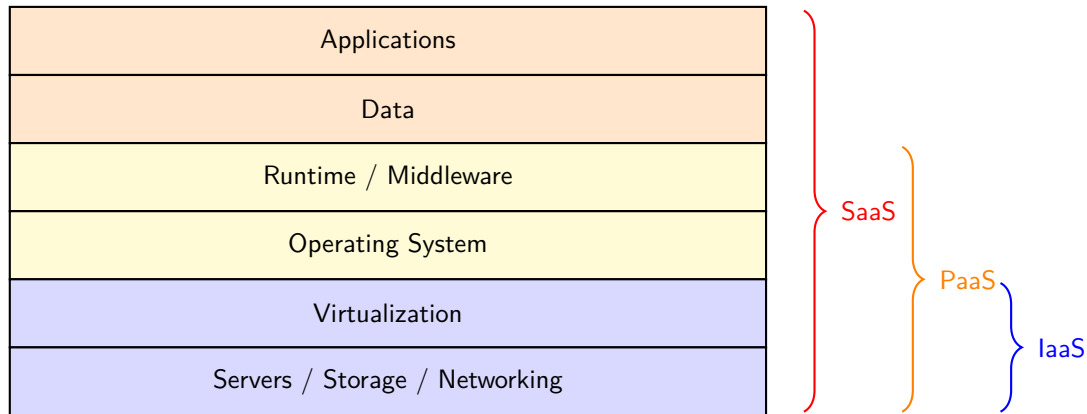
Characteristic	What it means	Real-world example
On-demand self-service	Provision via Application Programming Interface (API), no human interaction	AWS console: launch VM in 30 s
Broad network access	Available from any device over the network	Access S3 from laptop, phone, VM
Resource pooling	Shared infrastructure, multi-tenant	Same physical server, different tenants
Rapid elasticity	Scale up/down automatically	Black Friday: 10× more instances

Five Essential Characteristics

Characteristic	What it means	Real-world example
On-demand self-service	Provision via Application Programming Interface (API), no human interaction	AWS console: launch VM in 30 s
Broad network access	Available from any device over the network	Access S3 from laptop, phone, VM
Resource pooling	Shared infrastructure, multi-tenant	Same physical server, different tenants
Rapid elasticity	Scale up/down automatically	Black Friday: 10× more instances
Measured service	Pay only for what you use	Billed per CPU-hour, GB stored

All five must be present to qualify as “cloud computing”

Cloud Service Models – The Stack



IaaS, PaaS, SaaS: In Practice

IaaS, PaaS, SaaS: In Practice

IaaS (e.g. AWS EC2, Azure VMs, Google Compute Engine):

- Provider gives you VMs, storage, networks. You manage everything else
- “I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now”

IaaS, PaaS, SaaS: In Practice

IaaS (e.g. AWS EC2, Azure VMs, Google Compute Engine):

- Provider gives you VMs, storage, networks. You manage everything else
- “I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now”

PaaS (e.g. Google App Engine, AWS Elastic Beanstalk, Heroku):

- Provider gives you runtime + tools. You only manage your code and data
- “I wrote a Python web app. Just run it, I do not care about servers”

IaaS, PaaS, SaaS: In Practice

IaaS (e.g. AWS EC2, Azure VMs, Google Compute Engine):

- Provider gives you VMs, storage, networks. You manage everything else
- “I want a Linux server in Frankfurt, 4 CPUs, 16 GB RAM, right now”

PaaS (e.g. Google App Engine, AWS Elastic Beanstalk, Heroku):

- Provider gives you runtime + tools. You only manage your code and data
- “I wrote a Python web app. Just run it, I do not care about servers”

SaaS (e.g. Google Workspace, Microsoft 365, Slack, Zoom):

- Provider gives you a complete application. You manage your data only
- “I need email for 500 employees. I do not want to run a mail server”

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider

Service Models: Who Manages What?

Component	IaaS	PaaS	SaaS
Applications	You	You	Provider
Data	You	You	You
Runtime / Middleware	You	Provider	Provider
Operating System	You	Provider	Provider
Virtualization	Provider	Provider	Provider
Hardware	Provider	Provider	Provider
Network Services (LB, Proxy, VPN)	Provider	Provider	Provider

IaaS → SaaS: **less control, less responsibility.**

Major providers (AWS, Azure, GCP) offer all three models.

Discussion: Cloud Fundamentals

Case A: University

A large university wants a web tool for students to submit assignments. They have an IT department. Student data must comply with EU strict regulations.

Which model? Why?

- Control over data location needed
- Existing IT team can manage servers
- Ready-made products exist (Canvas, Moodle)
- No need to scale suddenly

Discussion: Cloud Fundamentals

Case A: University

A large university wants a web tool for students to submit assignments. They have an IT department. Student data must comply with EU strict regulations.

Which model? Why?

- Control over data location needed
- Existing IT team can manage servers
- Ready-made products exist (Canvas, Moodle)
- No need to scale suddenly

Case B: CloudBite

We are 3 engineers. We need to run a web app, store orders, and send email notifications. We must ship in 4 weeks and cannot afford to manage servers.

Which model? Why?

- No ops bandwidth: avoid IaaS if possible
- Web app: PaaS (just deploy the code)
- Email: SaaS (Mailchimp, SendGrid)
- Database: PaaS managed DB, or IaaS if we need fine-grained control

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms**
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

Cloud & Container Platforms

Private Cloud Platforms

- **Self-managed:** you own or rent the servers, install and manage the virtualization software yourself

Private Cloud Platforms

- **Self-managed:** you own or rent the servers, install and manage the virtualization software yourself
- Full control over hardware and data; requires dedicated staff and expertise

Private Cloud Platforms

- **Self-managed:** you own or rent the servers, install and manage the virtualization software yourself
- Full control over hardware and data; requires dedicated staff and expertise
- Typical use: regulated industries (banking, healthcare), large enterprises with existing data centers

Private Cloud Platforms

- **Self-managed:** you own or rent the servers, install and manage the virtualization software yourself
- Full control over hardware and data; requires dedicated staff and expertise
- Typical use: regulated industries (banking, healthcare), large enterprises with existing data centers

Private Cloud Platforms

- **Self-managed:** you own or rent the servers, install and manage the virtualization software yourself
- Full control over hardware and data; requires dedicated staff and expertise
- Typical use: regulated industries (banking, healthcare), large enterprises with existing data centers

Platform	Description	Logo
OpenStack	Open-source, widely used in private clouds	 openstack®
VMware vSphere	Enterprise leader, proprietary	
Proxmox VE	Open-source, gaining traction since recent VMware price rises	

Public Cloud Platforms

- **Managed / Public:** rent capacity on demand, pay-as-you-go

Public Cloud Platforms

- **Managed / Public:** rent capacity on demand, pay-as-you-go
- Provider handles hardware, networking, cooling, security, and updates

Public Cloud Platforms

- **Managed / Public:** rent capacity on demand, pay-as-you-go
- Provider handles hardware, networking, cooling, security, and updates

Public Cloud Platforms

- **Managed / Public:** rent capacity on demand, pay-as-you-go
- Provider handles hardware, networking, cooling, security, and updates

Provider	Description	Logo
Amazon Web Services (AWS)	Market leader since 2006	
Microsoft Azure	Strong enterprise integration	
Google Cloud Platform (GCP)	Data and AI focus	

These map directly to the IaaS model we just covered: the provider manages compute, storage, and networking.

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You
Virtualization	Provider	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You
Virtualization	Provider	You
Hardware	Provider	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You
Virtualization	Provider	You
Hardware	Provider	You
Networking	Provider	You

Public Cloud vs Private Cloud: What You Manage

Component	Public Cloud	Private Cloud
Applications	You	You
Data	You	You
Runtime / Middleware	You (IaaS) / Provider (PaaS)	You
Operating System	You (IaaS) / Provider (PaaS)	You
Virtualization	Provider	You
Hardware	Provider	You
Networking	Provider	You
Cooling / Power	Provider	You

Public cloud trades control for simplicity; private cloud gives full control at the cost of operational overhead.

Container Orchestration Platforms

- **Self-hosted / Private:** you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	

Container Orchestration Platforms

- **Self-hosted / Private:** you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	 docker
Kubernetes (K8s)	Created to solve containers running at scale: scheduling, scaling, networking, self-healing	 kubernetes

Container Orchestration Platforms

- **Self-hosted / Private:** you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	
Kubernetes (K8s)	Created to solve containers running at scale: scheduling, scaling, networking, self-healing	 kubernetes
OpenShift	Enterprise K8s by Red Hat; adds security, CI/CD, UI	 Red Hat OpenShift

Container Orchestration Platforms

- **Self-hosted / Private:** you install and operate on your own servers

Platform	Description	Logo
Docker	Container runtime; works well on a single host Managing hundreds of containers across hosts? Unmanageable	
Kubernetes (K8s)	Created to solve containers running at scale: scheduling, scaling, networking, self-healing	 kubernetes
OpenShift	Enterprise K8s by Red Hat; adds security, CI/CD, UI	 Red Hat OpenShift

Hosted / Public: same idea, managed by the cloud provider.

Examples: AWS ECS/EKS, Azure AKS, Google Kubernetes Engine (GKE), OpenShift on major clouds.

Discussion: Platforms

Our small startup Cloudbite needs to deploy a web application that may go from 100 to 100,000 users. Would you build a private cloud or use a public one? Why?

Think about:

- Upfront cost and time to market

Discussion: Platforms

Our small startup Cloudbite needs to deploy a web application that may go from 100 to 100,000 users. Would you build a private cloud or use a public one? Why?

Think about:

- Upfront cost and time to market
- Who manages infrastructure at night?

Discussion: Platforms

Our small startup Cloudbite needs to deploy a web application that may go from 100 to 100,000 users. Would you build a private cloud or use a public one? Why?

Think about:

- Upfront cost and time to market
- Who manages infrastructure at night?
- Can 3 engineers run a data center?

Discussion: Platforms

Our small startup Cloudbite needs to deploy a web application that may go from 100 to 100,000 users. Would you build a private cloud or use a public one? Why?

Think about:

- Upfront cost and time to market
- Who manages infrastructure at night?
- Can 3 engineers run a data center?
- What if traffic spikes overnight?

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture**
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)
 - **Subnets** (subdivisions of the address space)

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)
 - **Subnets** (subdivisions of the address space)
 - **Routing rules** and **security policies**

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)
 - **Subnets** (subdivisions of the address space)
 - **Routing rules** and **security policies**

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Networks in the Cloud

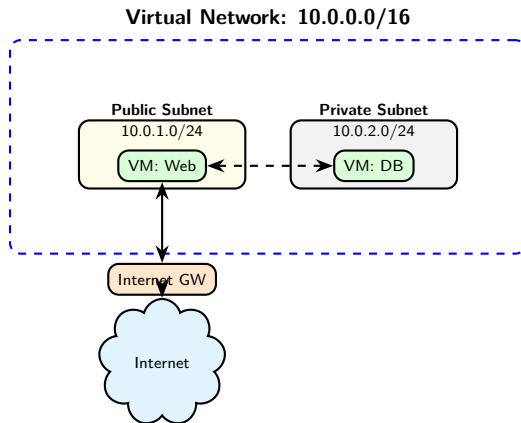
- Every cloud provider lets you create your own **isolated virtual network**
 - AWS: Virtual Private Cloud (VPC)
 - Azure: Virtual Network (VNet)
 - GCP: VPC Network
- Think of it as your **private data center network** in the cloud
- You define:
 - **Address space** (e.g., 10.0.0.0/16)
 - **Subnets** (subdivisions of the address space)
 - **Routing rules** and **security policies**

Key: under the hood, cloud providers use **overlay networks** (covered in Block 5) to isolate tenants at scale.

2

²AWS, "Amazon VPC User Guide"; Azure, "Virtual Network Documentation"; GCP, "VPC Documentation."

Virtual Network Architecture



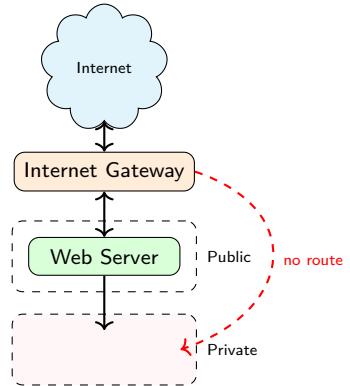
Subnets: Public vs Private

Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**

Private subnet:



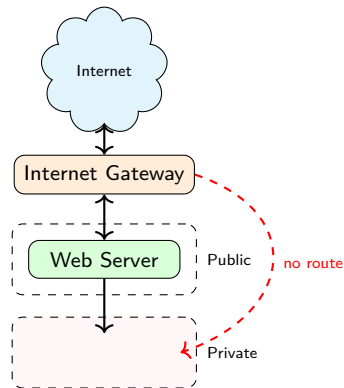
Subnets: Public vs Private

Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**

Private subnet:



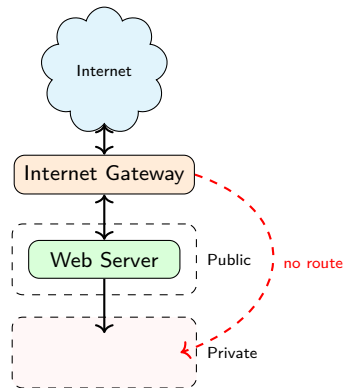
Subnets: Public vs Private

Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**
- Use: web servers, load balancers

Private subnet:



Subnets: Public vs Private

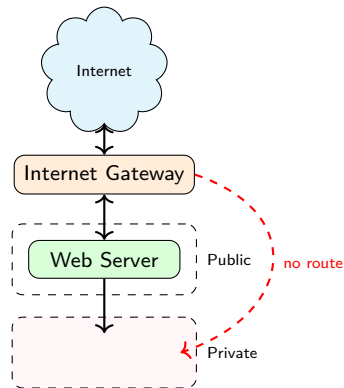
Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**
- Use: web servers, load balancers

Private subnet:

- **No route** to the Internet



Subnets: Public vs Private

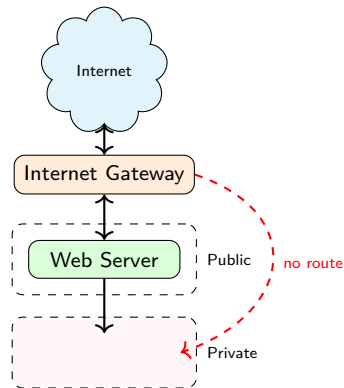
Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**
- Use: web servers, load balancers

Private subnet:

- **No route** to the Internet
- Only **private IPs**



Subnets: Public vs Private

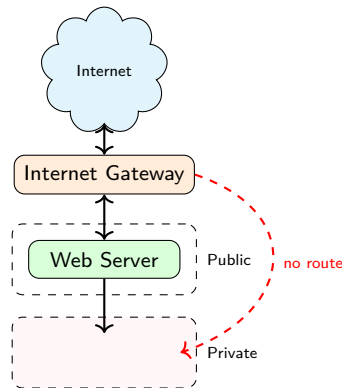
Cloud providers distinguish between subnets reachable from the Internet and isolated ones. Web servers need public access; databases and backends must stay hidden.

Public subnet:

- Route to an **Internet Gateway**
- Instances can have **public IPs**
- Use: web servers, load balancers

Private subnet:

- **No route** to the Internet
- Only **private IPs**
- Use: databases, backends

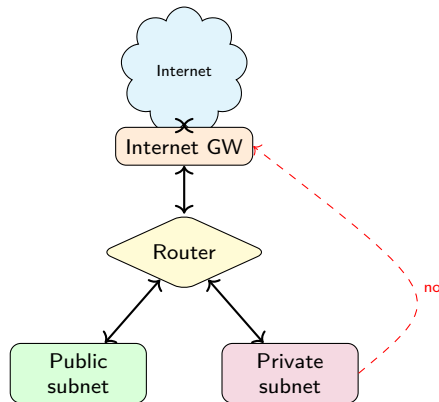


Route Tables

Every virtual network has an implicit **virtual router** managed by the provider. Route tables tell it where to forward traffic.

Dest.	Next Hop	Subnet
10.0.0.0/16	local	Pub + Priv
0.0.0.0/0	Internet GW	Public only

- **Local route:** traffic stays inside the virtual network

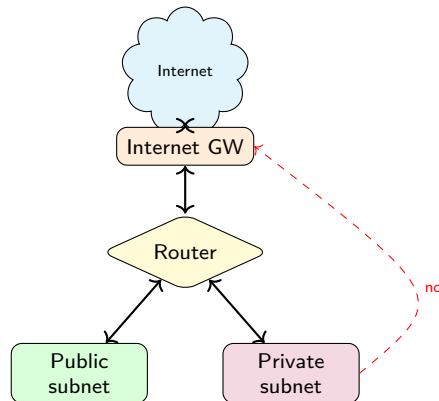


Route Tables

Every virtual network has an implicit **virtual router** managed by the provider. Route tables tell it where to forward traffic.

Dest.	Next Hop	Subnet
10.0.0.0/16	local	Pub + Priv
0.0.0.0/0	Internet GW	Public only

- **Local route:** traffic stays inside the virtual network
- **Default route** (0.0.0.0/0): public subnets exit via the Internet Gateway

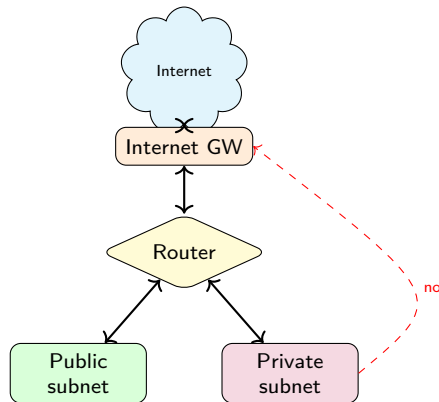


Route Tables

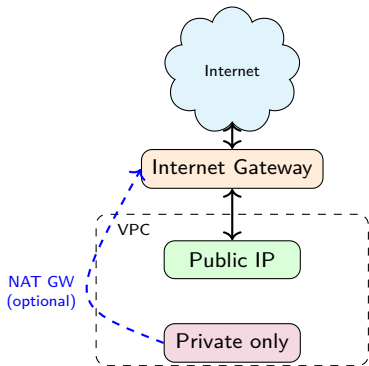
Every virtual network has an implicit **virtual router** managed by the provider. Route tables tell it where to forward traffic.

Dest.	Next Hop	Subnet
10.0.0.0/16	local	Pub + Priv
0.0.0.0/0	Internet GW	Public only

- **Local route:** traffic stays inside the virtual network
- **Default route** (0.0.0.0/0): public subnets exit via the Internet Gateway
- Private subnets have no default route



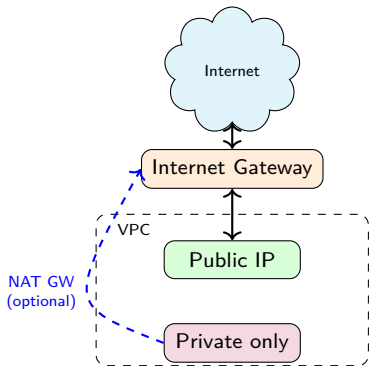
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside

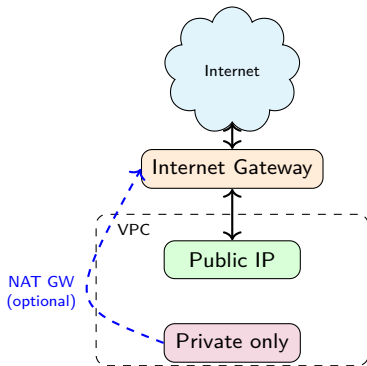
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently

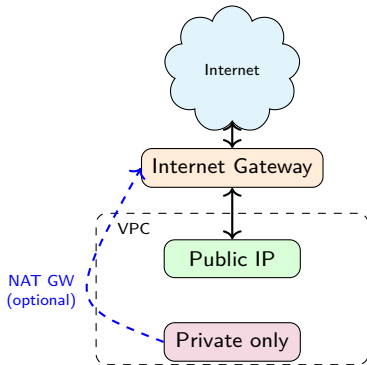
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently
- **Bidirectional:** instance is also reachable from the Internet

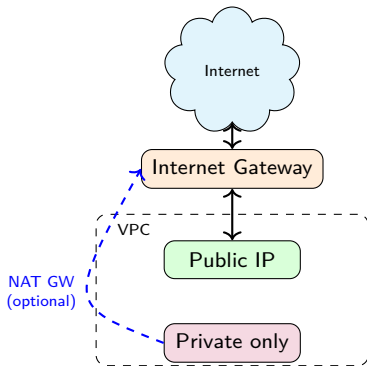
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently
- **Bidirectional:** instance is also reachable from the Internet
 - Requires a **public IP**

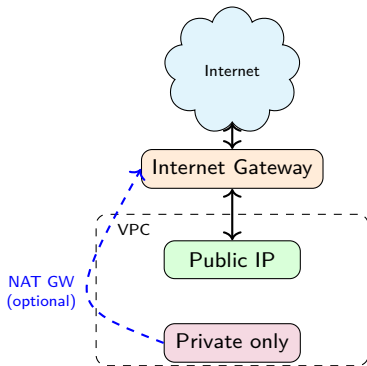
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently
- **Bidirectional:** instance is also reachable from the Internet
 - Requires a **public IP**
 - Use: web servers, load balancers

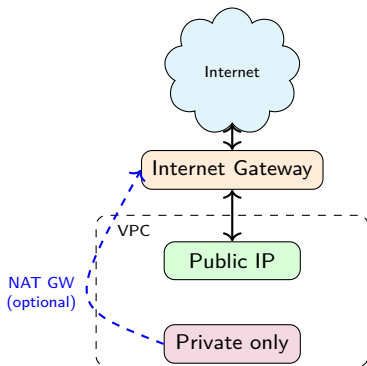
Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently
- **Bidirectional:** instance is also reachable from the Internet
 - Requires a **public IP**
 - Use: web servers, load balancers

Internet Gateway



In a cloud virtual network, instances are **isolated by default**. The Internet Gateway is the controlled entry and exit point.

- **Outbound only:** instance can reach the Internet, but is not reachable from outside
 - No public IP needed; the provider performs NAT transparently
- **Bidirectional:** instance is also reachable from the Internet
 - Requires a **public IP**
 - Use: web servers, load balancers

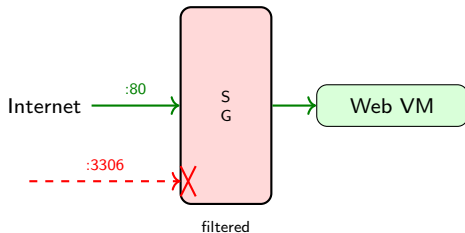
Note: a NAT Gateway can be added for private instances that need outbound Internet access (e.g. updates), without exposing them inbound.

Security Groups: Instance-Level Firewall

Once traffic reaches an instance, you still need to control what it can send and receive.

- **Security group** = virtual firewall on an instance
- Specifies allowed **inbound/outbound** traffic
- **Stateful**: allow inbound → response automatically allowed

Dir.	Proto	Port	Src/Dst
In	TCP	80	0.0.0.0/0
In	TCP	443	0.0.0.0/0
In	TCP	22	10.0.0.0/16
Out	All	All	0.0.0.0/0



By default: all inbound denied, all outbound allowed.

3

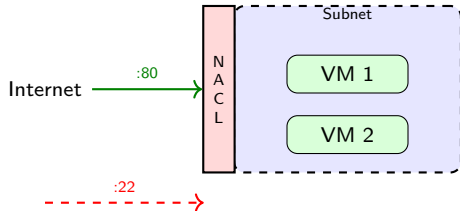
³AWS, "Security Groups for Your VPC," docs.aws.amazon.com.

Network ACLs (Access Control Lists): Subnet-Level Firewall

Network ACLs add a layer of control at the **subnet boundary**, before traffic reaches any instance.

- **Network ACL** = firewall at the **subnet** level
- Rules evaluated in **order** (numbered, first match wins)
- **Stateless**: must explicitly allow inbound AND outbound

#	Dir.	Proto	Port	Action
100	In	TCP	80	ALLOW
200	In	TCP	443	ALLOW
*	In	All	All	DENY



Filter applies to **all** instances in the subnet.

⁴AWS, "Network ACLs," docs.aws.amazon.com.

Security Groups vs Network ACLs

	Security Groups	Network ACLs
Scope	Instance level	Subnet level
State	Stateful	Stateless
Rules	Allow only	Allow + Deny
Evaluation	All rules evaluated	First match wins
Default	Deny all inbound	Allow all

Defense in depth: use both together.

- Security Groups → fine-grained per-instance control
- Network ACLs → broad subnet-level guardrails

Discussion: Cloud Networking

*We just deployed CloudBite on AWS.
Our web server must be reachable from
users, but the database must not be
exposed to the Internet. How do we set
this up?*

Think about:

- Our web server needs a public subnet with an Internet Gateway route

Discussion: Cloud Networking

*We just deployed CloudBite on AWS.
Our web server must be reachable from
users, but the database must not be
exposed to the Internet. How do we set
this up?*

Think about:

- Our web server needs a public subnet with an Internet Gateway route
- Our database goes in a private subnet: no direct route to the Internet

Discussion: Cloud Networking

*We just deployed CloudBite on AWS.
Our web server must be reachable from
users, but the database must not be
exposed to the Internet. How do we set
this up?*

Think about:

- Our web server needs a public subnet with an Internet Gateway route
- Our database goes in a private subnet: no direct route to the Internet
- Security group on the DB: allow traffic only from our web server

Discussion: Cloud Networking

*We just deployed CloudBite on AWS.
Our web server must be reachable from
users, but the database must not be
exposed to the Internet. How do we set
this up?*

Think about:

- Our web server needs a public subnet with an Internet Gateway route
- Our database goes in a private subnet: no direct route to the Internet
- Security group on the DB: allow traffic only from our web server
- NAT Gateway if the DB needs outbound access (e.g. pulling updates)

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services**
- 6 The Cloud Trade-Off
- 7 Session Summary

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

① Load Balancer (LB)

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

- 1 **Load Balancer (LB)**
- 2 **Proxy**

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

- 1 **Load Balancer (LB)**
- 2 **Proxy**
- 3 **Reverse Proxy**

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

- 1 Load Balancer (LB)
- 2 Proxy
- 3 Reverse Proxy
- 4 VPN (Virtual Private Network)

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

- 1 **Load Balancer (LB)**
- 2 **Proxy**
- 3 **Reverse Proxy**
- 4 **VPN (Virtual Private Network)**

Cloud Network Services: Overview

Beyond virtual networks and security controls, public cloud providers offer managed network services that simplify common infrastructure tasks, such as:

- 1 Load Balancer (LB)
- 2 Proxy
- 3 Reverse Proxy
- 4 VPN (Virtual Private Network)

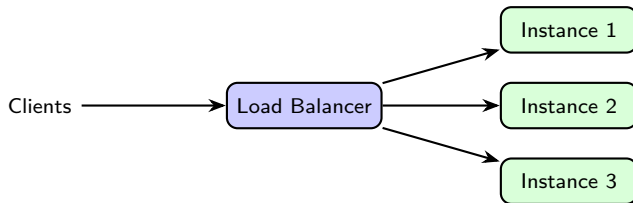
These services run as **managed offerings**: no servers to install or maintain.

Note: Proxy, Reverse Proxy, and VPN will be covered in depth in Block 6. Here we introduce them briefly.

Private cloud: these services are **not** provided automatically; you must deploy and manage them yourself (e.g. HAProxy for load balancing, OpenVPN for VPN).

Load Balancer

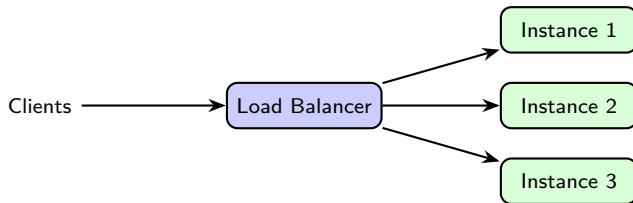
- Distributes incoming traffic across **multiple instances** (VMs or containers)



⁵AWS, "Elastic Load Balancing Documentation," docs.aws.amazon.com.

Load Balancer

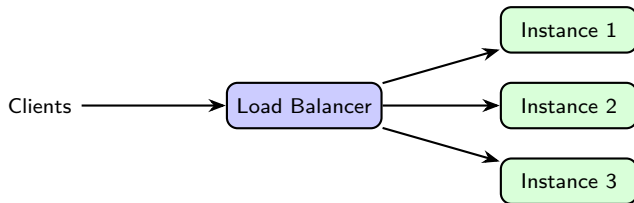
- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed



⁵AWS, "Elastic Load Balancing Documentation," docs.aws.amazon.com.

Load Balancer

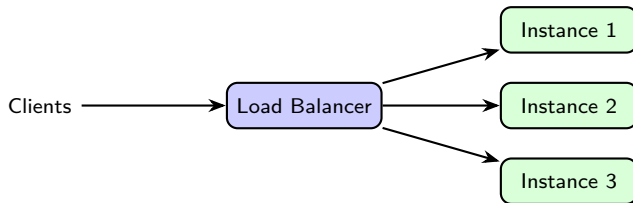
- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed
- If one instance fails, traffic is **redirected** to healthy ones



⁵AWS, "Elastic Load Balancing Documentation," docs.aws.amazon.com.

Load Balancer

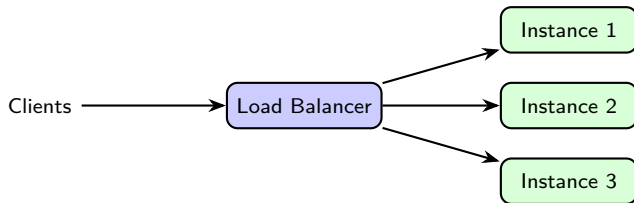
- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed
- If one instance fails, traffic is **redirected** to healthy ones



⁵AWS, "Elastic Load Balancing Documentation," docs.aws.amazon.com.

Load Balancer

- Distributes incoming traffic across **multiple instances** (VMs or containers)
- Prevents any single instance from being overwhelmed
- If one instance fails, traffic is **redirected** to healthy ones

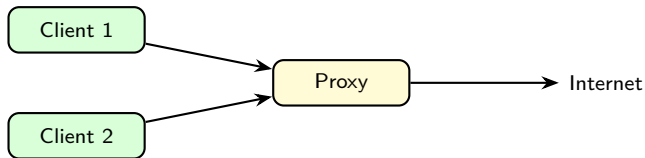


- **L4 (Transport Layer)** Load Balancer: routes based on IP and port (fast, simple)
- **L7 (Application Layer)** Load Balancer: routes based on HTTP headers, URL path, cookies (smarter)
- Cloud examples: AWS ALB/NLB, Azure Load Balancer, GCP Cloud Load Balancing

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

- Clients send requests to the proxy, which forwards them to the Internet

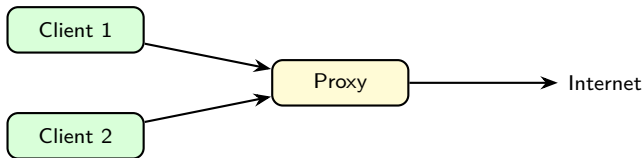


- **Access control:** block certain websites or domains

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

- Clients send requests to the proxy, which forwards them to the Internet

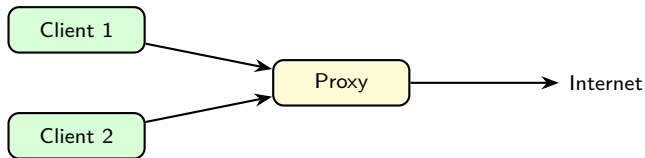


- **Access control:** block certain websites or domains
 - Security: block known malicious domains

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

- Clients send requests to the proxy, which forwards them to the Internet

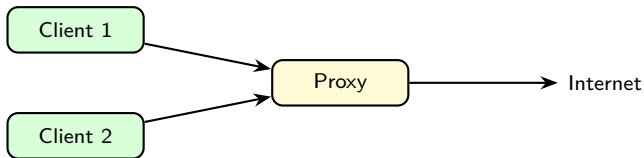


- **Access control:** block certain websites or domains
 - Security: block known malicious domains
 - Company policy: restrict access to non-work-related content

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

- Clients send requests to the proxy, which forwards them to the Internet

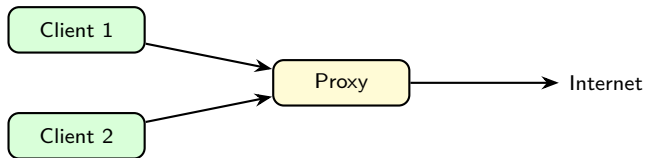


- **Access control:** block certain websites or domains
 - Security: block known malicious domains
 - Company policy: restrict access to non-work-related content
- **Caching:** store frequently accessed content, reduce bandwidth

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

- Clients send requests to the proxy, which forwards them to the Internet

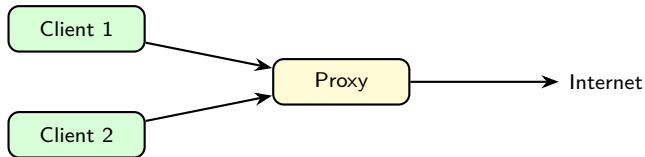


- **Access control:** block certain websites or domains
 - Security: block known malicious domains
 - Company policy: restrict access to non-work-related content
- **Caching:** store frequently accessed content, reduce bandwidth
- **Logging:** monitor what employees or VMs access

Proxy

Without a proxy, every instance in your network reaches the Internet directly: no visibility, no control. A proxy sits in front of clients and mediates all outbound traffic.

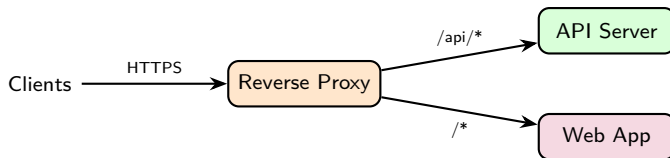
- Clients send requests to the proxy, which forwards them to the Internet



- **Access control:** block certain websites or domains
 - Security: block known malicious domains
 - Company policy: restrict access to non-work-related content
- **Caching:** store frequently accessed content, reduce bandwidth
- **Logging:** monitor what employees or VMs access
- Examples: Squid, corporate firewalls with proxy mode

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

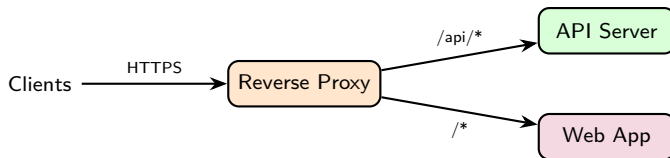


- **TLS termination:** handles HTTPS, backends receive plain HTTP

Will be covered in depth in Block 6.

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

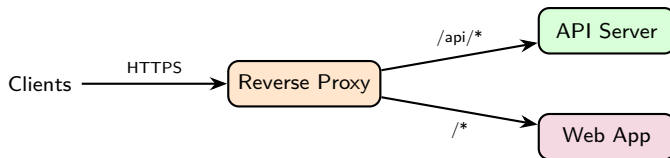


- **TLS termination:** handles HTTPS, backends receive plain HTTP
- **URL routing:** directs requests to the right backend service

Will be covered in depth in Block 6.

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

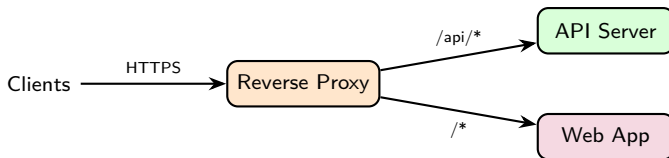


- **TLS termination:** handles HTTPS, backends receive plain HTTP
- **URL routing:** directs requests to the right backend service
- **Hides backend servers:** clients only see the proxy's address

Will be covered in depth in Block 6.

Reverse Proxy

A reverse proxy sits in front of backend servers and mediates all inbound traffic. Clients never communicate directly with the backend.

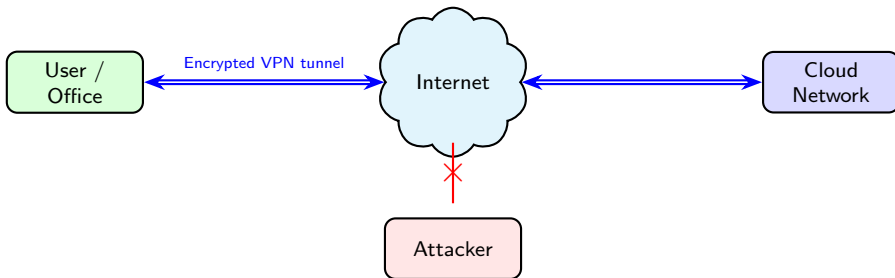


- **TLS termination:** handles HTTPS, backends receive plain HTTP
- **URL routing:** directs requests to the right backend service
- **Hides backend servers:** clients only see the proxy's address
- Examples: NGINX, HAProxy, AWS CloudFront, Azure Front Door

Will be covered in depth in Block 6.

VPN: Virtual Private Network

- A **VPN (Virtual Private Network)** creates a **secure, encrypted tunnel** over the public Internet
- Connects remote networks or users as if they were on the same private network



Two main types:

- **Site-to-site VPN:** office ↔ cloud network
- **Client VPN:** one user ↔ remote network
- Examples: AWS VPN, Azure VPN Gateway, GCP Cloud VPN

Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)

Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)
- Reverse proxy to route `/api/*` to API server

Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)
- Reverse proxy to route `/api/*` to API server
- Database in private subnet, no public IP

Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)
- Reverse proxy to route `/api/*` to API server
- Database in private subnet, no public IP
- Site-to-site VPN for office access

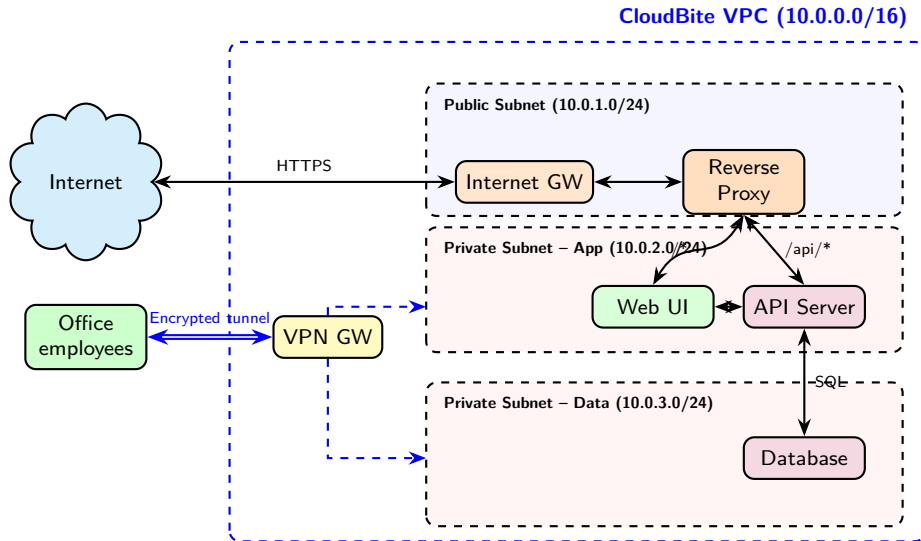
Discussion: Cloud Network Services

A company has a public frontend, a private API, and a database. Employees also need to access the cloud from the office. Which services would you use?

Think about:

- Load balancer in front of frontend (public)
- Reverse proxy to route `/api/*` to API server
- Database in private subnet, no public IP
- Site-to-site VPN for office access
- Security groups + ACLs at each tier

CloudBite Architecture: Putting It All Together



Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off**
- 7 Session Summary

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising
- The team now has 30 engineers; we *could* run our own servers

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising
- The team now has 30 engineers; we *could* run our own servers
- We now ask: “Should we leave the cloud?”

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising
- The team now has 30 engineers; we *could* run our own servers
- We now ask: “Should we leave the cloud?”

CloudBite, Five Years Later

- We grew to 2 million users. AWS bill: **\$40,000/month** and rising
- The team now has 30 engineers; we *could* run our own servers
- We now ask: “Should we leave the cloud?”

The cloud is not always the cheapest option. Let us explore why.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years
- In 2023, they **moved everything to their own hardware**.

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on *AWS* for 15 years
- In 2023, they **moved everything to their own hardware**.
- Result: **\$10 million saved over five years** (50%+ cost reduction)

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on *AWS* for 15 years
- In 2023, they **moved everything to their own hardware**.
- Result: **\$10 million saved over five years** (50%+ cost reduction)

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years
- In 2023, they **moved everything to their own hardware**.
- Result: **\$10 million saved over five years** (50%+ cost reduction)

Why it worked for them:

- **Stable, predictable workload** → no sudden traffic spikes
- **Large, experienced ops team** → already managing servers regardless
- **No need for global presence** → most users in the same region

⁶37signals, "We left the cloud," basecamp.com/cloud-exit, 2023.

Case Study: Basecamp Leaves the Cloud

- **Basecamp / 37signals** (creators of Ruby on Rails) ran on AWS for 15 years
- In 2023, they **moved everything to their own hardware**.
- Result: **\$10 million saved over five years** (50%+ cost reduction)

Why it worked for them:

- **Stable, predictable workload** → no sudden traffic spikes
- **Large, experienced ops team** → already managing servers regardless
- **No need for global presence** → most users in the same region

Key insight

Basecamp's exit worked because their workload was **predictable**. For a startup with unpredictable traffic, the cloud's elasticity remains essential.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:
 - AWS charges \$0.09/GB for outbound traffic

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:
 - AWS charges \$0.09/GB for outbound traffic
 - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:
 - AWS charges \$0.09/GB for outbound traffic
 - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone
 - Egress fees increased 20% in 2023; inter-Availability-Zone (AZ) fees doubled in 2025

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:
 - AWS charges \$0.09/GB for outbound traffic
 - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone
 - Egress fees increased 20% in 2023; inter-Availability-Zone (AZ) fees doubled in 2025

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

Vendor Lock-In: The Hidden Cost

- Cloud providers design services that are **easy to adopt, hard to leave**
- Proprietary services create **architectural dependencies**:
 - AWS Lambda, DynamoDB, SQS → no direct equivalent on Azure or GCP
 - Rewriting the application can cost more than years of cloud bills
- **Data egress fees** make leaving expensive:
 - AWS charges \$0.09/GB for outbound traffic
 - Moving 50 TB out of AWS $\approx$ \$4,500 in fees alone
 - Egress fees increased 20% in 2023; inter-Availability-Zone (AZ) fees doubled in 2025
- **68% of enterprises** exceeded their cloud budget due to unexpected data transfer costs (Flexera, 2024)

7

⁷Flexera, "State of the Cloud Report," 2024; AWS pricing documentation, 2025.

The Pricing Ratchet

- The commercial cloud boom started around **2006**: Amazon S3 and Elastic Compute Cloud (EC2) made infrastructure rentable by API
- Adoption accelerated in **2008 to 2012**: startups could launch without buying servers first
- Providers reduced entry friction with **free tiers**, credits, and pay-as-you-go pricing

The Pricing Ratchet

- The commercial cloud boom started around **2006**: Amazon S3 and Elastic Compute Cloud (EC2) made infrastructure rentable by API
- Adoption accelerated in **2008 to 2012**: startups could launch without buying servers first
- Providers reduced entry friction with **free tiers**, credits, and pay-as-you-go pricing
- Once adoption deepens, dependency grows:
 - Data migration takes time and money
 - Proprietary services increase lock-in
 - Re-architecting may cost more than staying

The Pricing Ratchet

- The commercial cloud boom started around **2006**: Amazon S3 and Elastic Compute Cloud (EC2) made infrastructure rentable by API
- Adoption accelerated in **2008 to 2012**: startups could launch without buying servers first
- Providers reduced entry friction with **free tiers**, credits, and pay-as-you-go pricing
- Once adoption deepens, dependency grows:
 - Data migration takes time and money
 - Proprietary services increase lock-in
 - Re-architecting may cost more than staying
- This pattern appears across industries:
 - **Video streaming**: low initial pricing to gain users, then gradual increases
 - **Cloud computing**: same logic, but switching costs are higher because the whole architecture depends on the provider

The pattern

Low entry cost → fast adoption → architectural dependency → weak bargaining power.

So... Cloud or Not?

	Cloud (public)	On-premises / private
Best when	Unpredictable traffic, small team	Stable workload, large ops team
Cost model	OpEx (pay-as-you-go)	CapEx (buy hardware upfront)
Scaling	Instant, global	Weeks to months (buy + install)
Risk	Vendor lock-in, price hikes	Hardware failures, staffing
Control	Limited (provider's rules)	Full (your hardware, your rules)

- **Most real-world deployments are hybrid:** critical/stable workloads on-premises, burst/variable workloads in the cloud
- There is no universal answer; **every case is a trade-off**

Outline

- 1 From Virtualization to Cloud
- 2 Fundamentals of Cloud Computing
- 3 Cloud & Container Platforms
- 4 Cloud Network Architecture
- 5 Cloud Network Services
- 6 The Cloud Trade-Off
- 7 Session Summary**

Key Takeaways: CloudBite's Journey

What we learned:

- ① Virtualization → cloud: technology vs **business model**

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- ① Virtualization → cloud: technology vs **business model**
- ② Five NIST characteristics define true cloud

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- ① Virtualization → cloud: technology vs **business model**
- ② Five NIST characteristics define true cloud
- ③ **IaaS / PaaS / SaaS**: who manages what

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- ① Virtualization → cloud: technology vs **business model**
- ② Five NIST characteristics define true cloud
- ③ **IaaS / PaaS / SaaS**: who manages what
- ④ Public vs Private cloud.

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- ① Virtualization → cloud: technology vs **business model**
- ② Five NIST characteristics define true cloud
- ③ **IaaS / PaaS / SaaS**: who manages what
- ④ Public vs Private cloud.
- ⑤ Designed a **virtual network** with public and private subnets

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive
- 10 Prices increase once you depend on the provider

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive
- 10 Prices increase once you depend on the provider
- 11 On-premises works for **stable**, **predictable** workloads

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive
- 10 Prices increase once you depend on the provider
- 11 On-premises works for **stable**, **predictable** workloads
- 12 **Hybrid** is often the pragmatic answer

Key Takeaways: CloudBite's Journey

What we learned:

- 1 Virtualization → cloud: technology vs **business model**
- 2 Five NIST characteristics define true cloud
- 3 **IaaS / PaaS / SaaS**: who manages what
- 4 Public vs Private cloud.
- 5 Designed a **virtual network** with public and private subnets
- 6 Secured it with **security groups** + **ACLs**
- 7 Added a **load balancer**, **reverse proxy**, **VPN**

What we learned:

- 8 The cloud is **not free**: costs grow with scale
- 9 **Vendor lock-in** makes leaving expensive
- 10 Prices increase once you depend on the provider
- 11 On-premises works for **stable, predictable** workloads
- 12 **Hybrid** is often the pragmatic answer
- 13 **Every case is a trade-off**; there is no universal solution

References

- ❶ NIST SP 800-145, Mell & Grance, “The NIST Definition of Cloud Computing,” 2011.
- ❷ AWS, “Amazon VPC User Guide,” docs.aws.amazon.com.
- ❸ Microsoft Azure, “Azure Virtual Network Documentation,” docs.microsoft.com.
- ❹ Google Cloud, “VPC Documentation,” cloud.google.com/vpc/docs.
- ❺ CSA, “Security Guidance for Critical Areas of Focus in Cloud Computing v5,” 2022.
- ❻ Erl, Puttini & Mahmood, *Cloud Computing: Concepts, Technology & Architecture*, Prentice Hall, 2013.
- ❼ Kurose & Ross, *Computer Networking: A Top-Down Approach*, 8th ed., Pearson, 2021.
- ❽ AWS, “Elastic Load Balancing Documentation,” docs.aws.amazon.com.
- ❾ NGINX, Inc., “What Is a Reverse Proxy?,” nginx.com/resources.
- ❿ AWS, “AWS VPN Documentation,” docs.aws.amazon.com.
- ⓫ AWS, “Amazon EC2 FAQs” and “AWS Free Tier,” aws.amazon.com.
- ⓬ Microsoft Azure, “Free Azure services,” azure.microsoft.com.
- ⓭ Google Cloud, “Free Trial and Free Tier Services,” cloud.google.com.